



RACEnergy Founders Arun and Gautham with their battery pack

Electronics For You. In absolute numbers, four of these batteries allow an electric three-wheeler (E3W) with a range of about 110km. Reddy adds that it takes about 30 to 60 seconds to swap these batteries. This results in E3Ws spending more time on the road, a prerequisite for a sustainable business for E3W owners and fleet operators.

Weighing around 9.5kg each, these batteries feature an IP68 rating. Reddy explains that the process of developing these batteries was the toughest part of the start-up's journey that started in 2019. Their initial designs included a plastic enclosure which was replaced with aluminium because of the latter's heat dissipating properties. To ensure the designed batteries work with the software built in-house, the startup also had to design its own battery management system (BMS).

Reddy highlights, "BMS sourced

from third parties was not an ideal match for our batteries, so we worked on the same ourselves. Even the cells inside the battery feature immersion technology which dissipates the heat to aluminium. It has been hard because the more density you pack inside a battery, the more sensitive it becomes."

Apart from cells and chips, the startup has absolute control over the components and raw materials it sources for manufacturing batteries and battery swap stations. Reddy says his startup is focusing on the B2B space as most of the B2C electric vehicle users commute for a limited number of kilometres every day, and it is the B2B space that needs major support in finding sustainable modes of transport.

"At present, we are conducting about 650 battery swaps just out of Hyderabad every day. I can't put a number on it, but before the end of this fiscal year, this figure would be

in the thousands," he says.

Having recently raised \$3 million in a pre-series A round, RACEnergy has commissioned its second battery manufacturing facility with the capability to manufacture close to 30,000 batteries per year. For the record, the startup has so far raised a total of \$6 million from strategic investors, including growx Ventures, Micelio Mobility, Huddle, and other angel investors.

Primarily focused on the E3W vertical, RACEnergy has also forayed into the electric two-wheeler space. Currently, 80% of the battery swaps happening on the RACEnergy platform are for E3Ws. Reddy shares that, apart from the fleet operators the startup is working with, it is also in the process of announcing collaborations with vehicle OEMs.

"We are in final discussions with three major OEMs and are hopeful of announcing these in the coming months," he shares. Additionally, RACEnergy and Hala Mobility have announced their collaboration to deploy 2,000 electric vehicles in India.

Operating on different business models that include setting up swap stations on its own and via the franchise route, RACEnergy is now looking to expand beyond Hyderabad. Reddy says that while it will deploy battery swap solutions before the end of this fiscal year, it will also cover tier 3 and tier 4 geographies near Hyderabad. In numbers, RACEnergy has a goal of targeting 80,000 two- and three-wheelers by 2025 through 250,000 swappable batteries in circulation before the end of 2025.

Last-mile logistics and transportation have become the biggest enablers of EV adoption in India. Battery swap, according to various reports, is encouraging more and more commercial drivers to put faith in electric powertrains. **EFY**

—Mukul Yudhveer Singh